

Martinez, Thomas - Research Proposal

1. INTRODUCTION

My research area is **algebraic combinatorics**, with a focus on the geometry of flag varieties, cluster algebras, and link invariants. I develop diagrammatic models for spaces of *flags*, nested sequences of vector subspaces. The algebraic models are *cluster algebras*, generated by collections of variables related by explicit exchange rules. The topological models are *links*, collections of disjoint circles in the three-sphere, studied through invariants unchanged by deformations through embeddings.

Recent work establishes cluster structures on *braid varieties*, spaces of flag sequences with successive relative positions prescribed by crossings of a braid [1–3]. Galashin–Lam relate link invariants to Richardson point counts and cohomology, graded vector spaces recording topology [4]. With Roger Casals at UC Davis, I propose to develop affine counterparts of these connections. My objectives are to construct **diagrammatic cluster coordinates** generating regular functions and to recover **geometric invariants through link topology**.

2. BACKGROUND

An *open Richardson variety* consists of flags with prescribed intersection dimensions relative to two fixed flags in general position. In the type-A setting, affine flags are lattice chains $L_i \supset L_{i+1}$ in $\mathbb{C}((t))^n$ with $\dim_{\mathbb{C}}(L_i/L_{i+1}) = 1$ and $L_{i+n} = tL_i$: each lattice is a free rank- n module over $\mathbb{C}[[t]]$ spanning this Laurent-series space. Their open Richardson varieties $\mathring{R}_{u,w}$ are finite-dimensional, with u, w encoding relative positions.

Concrete examples come from the *Grassmannian* $\mathrm{Gr}(k, n)$, the space of k -dimensional subspaces of $\mathbb{C}^n$. Representing a subspace by the rows of a full-rank matrix gives homogeneous *Plücker coordinates* Δ_I , its maximal minors. *Open positroid varieties* Π_f° impose prescribed ranks on cyclic intervals of columns, a construction originating in the study of nonnegative matrix minors [5, 6]. Their cluster structures extend those of Grassmannians [7, 8]. Snider identifies each nonempty patch where a minor is nonzero with an affine open Richardson variety [9]:

$$\Pi_f^\circ \cap \{\Delta_I \neq 0\} \cong \mathring{R}_{f, t_I},$$

where f indexes the positroid and t_I is the translation specifying the chart.

A *seed* Σ consists of algebraically independent rational functions and a *quiver*, a directed graph encoding their exchange rules. *Mutation* replaces one variable and updates the quiver; *frozen variables* are never mutated. The cluster algebra $\mathcal{A}$ is generated by variables from all seeds, with frozen variables inverted here. Its *upper cluster algebra* $\mathcal{U}$ consists of functions Laurent in every seed.

My **affine Deodhar diagrams**, periodic strand diagrams made of crossings and elbows, encode decompositions into pieces $(\mathbb{C}^*)^{e(D)} \times \mathbb{C}^{b(D)}$ with nonzero and unrestricted coordinates [10]. An *algebraic torus* is a product of copies of $\mathbb{C}^*$.

Gorsky–Kim–Sherman–Bennett classify algebraic tori among finite type-A open Richardson varieties [11, Theorem 1.2]; Bossinger–Gorsky–Simental extend this classification to finite crystallographic types [12, Theorem 1.8]. In joint work with Matthew J. Tyler, I establish an affine analogue, including the following criterion [13, Theorem 1.4].

Theorem 1 (Martinez–Tyler [13]). *An open Richardson variety in affine type A is an algebraic torus if and only if its affine Deodhar decomposition consists of a single stratum.*

For the standard torus action, Tsukerman–Williams and Can–Saha give combinatorial criteria for dense orbits on finite Richardson varieties [14, 15]. With Tyler, I extend these criteria to type-A braid Richardson varieties and identify their topological meaning through the following result [13, Theorem 1.1]. A link is *smoothly slice* if its components bound disjoint smooth disks in the four-ball.

Theorem 2 (Martinez–Tyler [13]). *A braid Richardson variety in finite type A has a dense orbit under its standard torus action if and only if its associated link is smoothly slice.*

The second criterion concerns the standard torus action; the first concerns the variety’s abstract isomorphism type.

3. RESEARCH PLAN

3.1. Geometric cluster structures. Bao–Ye identify Richardson coordinate rings, consisting of functions with no poles on the variety, with upper cluster algebras and conjecture **local acyclicity** [16, Conjecture 5.15]: a finite cover by cluster localizations, where inverting selected variables agrees with freezing them, and each resulting algebra has a seed without directed cycles among mutable vertices. I will identify our diagrammatic seeds with theirs and prove local acyclicity for the matched affine families. Muller’s theorem would then yield [17]

$$\mathbb{C}[\mathring{R}_{u,w}] = \mathcal{A}(\Sigma_{u,w}) = \mathcal{U}(\Sigma_{u,w}),$$

showing that cluster variables generate every regular function.

Joint work with Moriah Elkin and Tyler constructs candidate quivers from *affine three-dimensional plabic diagrams*, periodic spatial graphs with black and white vertices, and proves local acyclicity of their cluster algebras. I will compare their coordinate functions and exchange rules with Bao–Ye’s construction, beginning with tori and one-mutable-variable seeds. Finite-type comparisons provide a model [1–3]; their use of a longest Weyl-group element, a maximal-length relative position, requires new arguments in affine type.

For patches, *weak separation* gives accessible examples: I, J are weakly separated if no cyclically ordered a, b, c, d satisfy $a, c \in I \setminus J$ and $b, d \in J \setminus I$ [18]. Our joint work equates disjointness of diagrammatic surfaces with weak separation of I from every member of the *Grassmann necklace*, the lexicographically minimal positroid bases for each cyclic order. I will compare these seeds with *freezing* a face variable of a planar plabic graph, forbidding its mutation and inverting it, then analyze intersecting configurations.

3.2. Link invariants and cohomology. The *HOMFLY polynomial* $P_L(a, q)$ of a link L is a two-variable link invariant determined by local crossing relations. *Khovanov–Rozansky homology* refines it to graded vector spaces whose graded Euler characteristic recovers the polynomial. In finite type, a normalized HOMFLY coefficient computes Richardson point counts, while a specified part of this homology recovers torus-equivariant compactly supported Richardson cohomology, incorporating the torus action [4]. I will seek an **affine point-count formula** by closing periodic braid strands and matching crossing relations with Deodhar recursions:

$$H_{u,w}^{\text{top}}(q) = \# \mathring{R}_{u,w}(\mathbb{F}_q) = \sum_D (q-1)^{e(D)} q^{b(D)}.$$

Here the middle term counts solutions over the q -element field, H^{top} is a normalized top a -degree coefficient, and D indexes the diagrammatic pieces. Existing solid-torus link invariants provide models [19]. I will establish closure invariance and normalization, starting with tori and short periodic braids. My computations verify finite-type specializations for braids associated with permutations of six letters.

A *mixed Hodge structure* equips cohomology with filtrations reflecting algebraic geometry. My longer-term goal is to recover the graded pieces of its weight filtration from link homology. Lam–Speyer’s Lefschetz results impose symmetries on these cohomological data [20]. I will investigate

whether the proposed affine cluster structures have full-rank extended exchange matrices and satisfy the *Louise property*, a recursive covering condition obtained by freezing and inverting cluster variables, so that their results apply.

Casals's joint work realizes open positroid varieties as *augmentation varieties*, spaces of scalar solutions to equations from *Legendrian links*, links tangent to a nonintegrable plane field [21]. With Gao, he realizes cluster seeds by *Lagrangian fillings*, surfaces bounding the link on which the ambient symplectic form vanishes, for positive braids [22]. I will investigate how these constructions extend to my periodic diagrams.

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